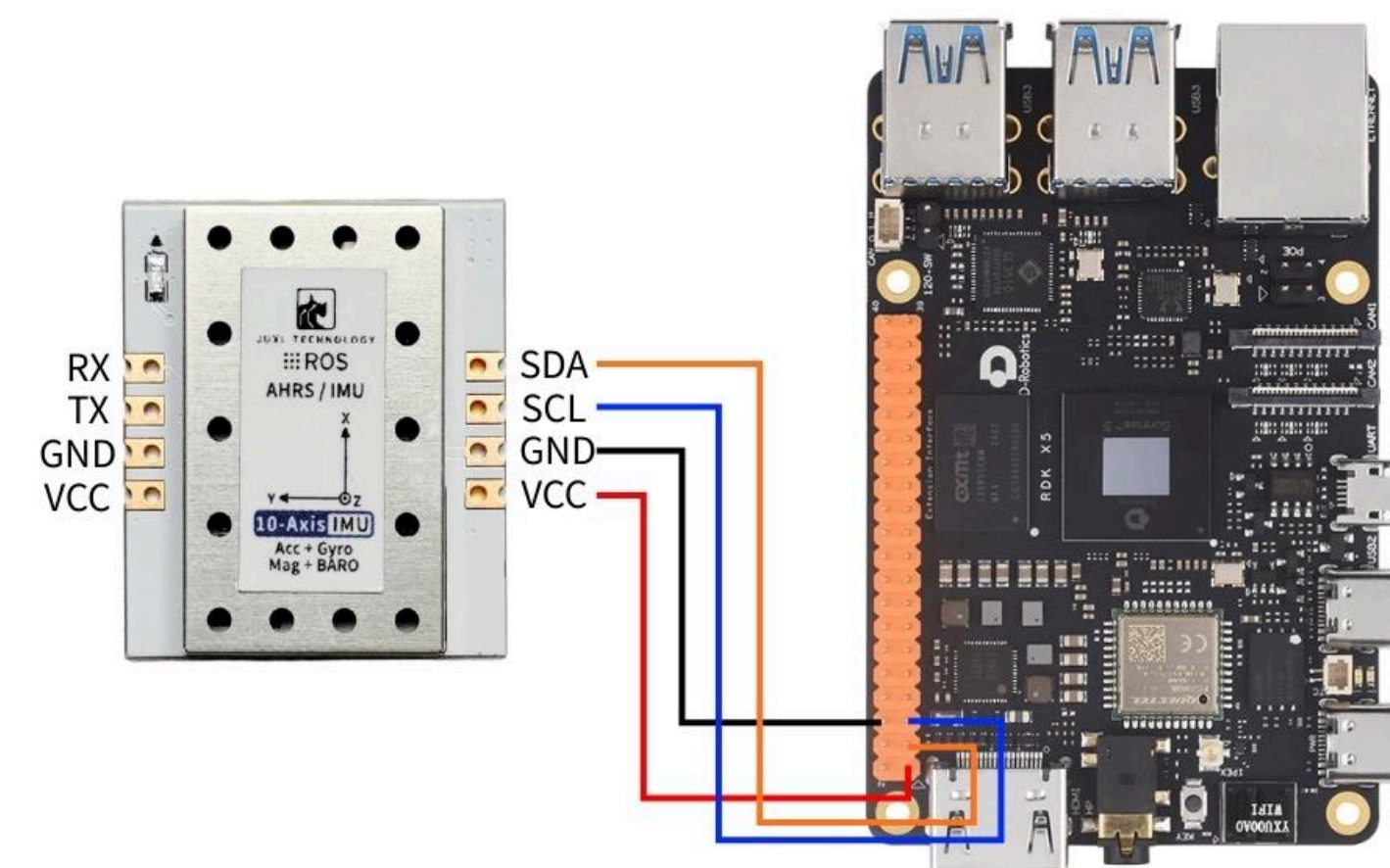


RDK Series

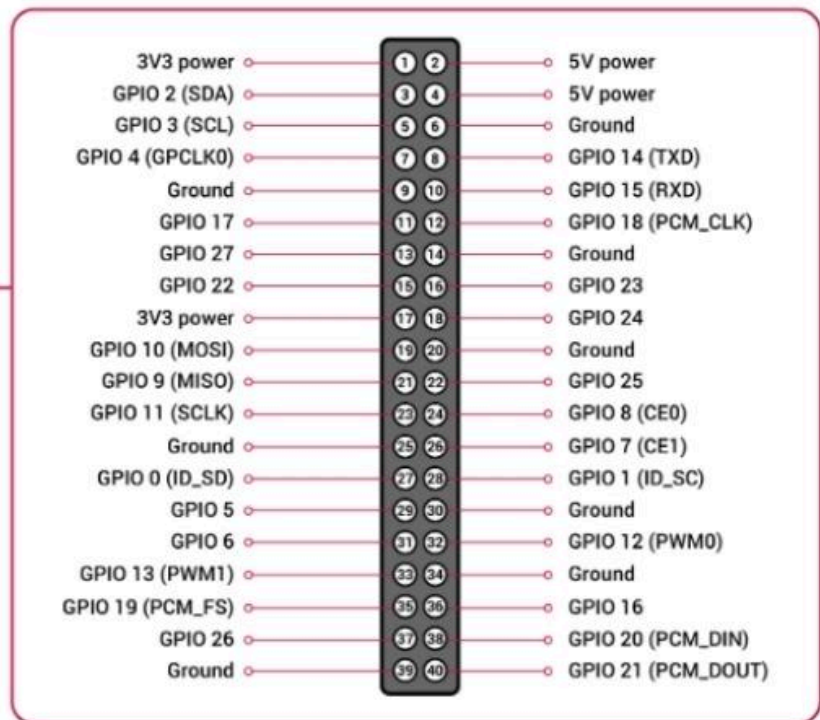
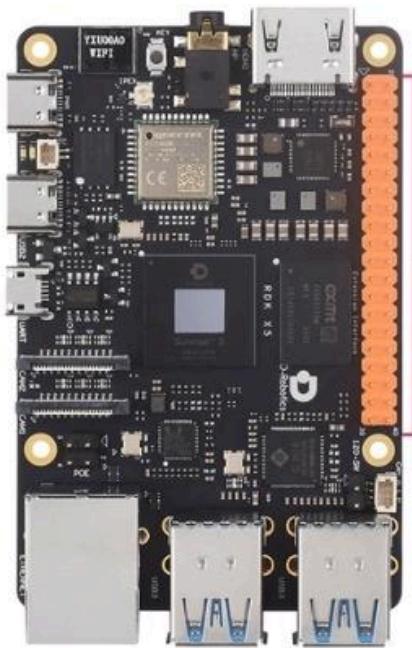
Step 1 Connect devices

This tutorial takes the mirroring of the? version of the RDK X5 motherboard as an example.

Connect the IMU attitude sensor to the I2C interface of the RDK X5 as shown below.



IMU姿态传感器	RDK5（物理引脚）
SDA	3
SCL	5
GND	GND
3.3V	3.3V



2. Check device status

View I2C Devices

代码块

```
1 python3 /app/40pin_samples/test_i2c.py
```

```
sunrise@ubuntu:~$ python3 /app/40pin_samples/test_i2c.py
Starting demo now! Press CTRL+C to exit
List of enabled I2C controllers:
/dev/i2c-0 /dev/i2c-2 /dev/i2c-3 /dev/i2c-4 /dev/i2c-5 /dev/i2c-6 /dev/i2c-7 /dev/i2c-8
Please input I2C BUS num 5
  0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
10:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
20:  --  --  --  23  --  --  --  --  --  --  --  --  --  --  --  --
30:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
40:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
50:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
60:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
70:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
```

3. Install the driver library

3.1 Install the Python libraries required for the code

代码块

```
1 sudo apt update
2 sudo apt install -y python3-serial
```

```
3 sudo apt install -y python3-smbus2
```

3.2 Transfer Files



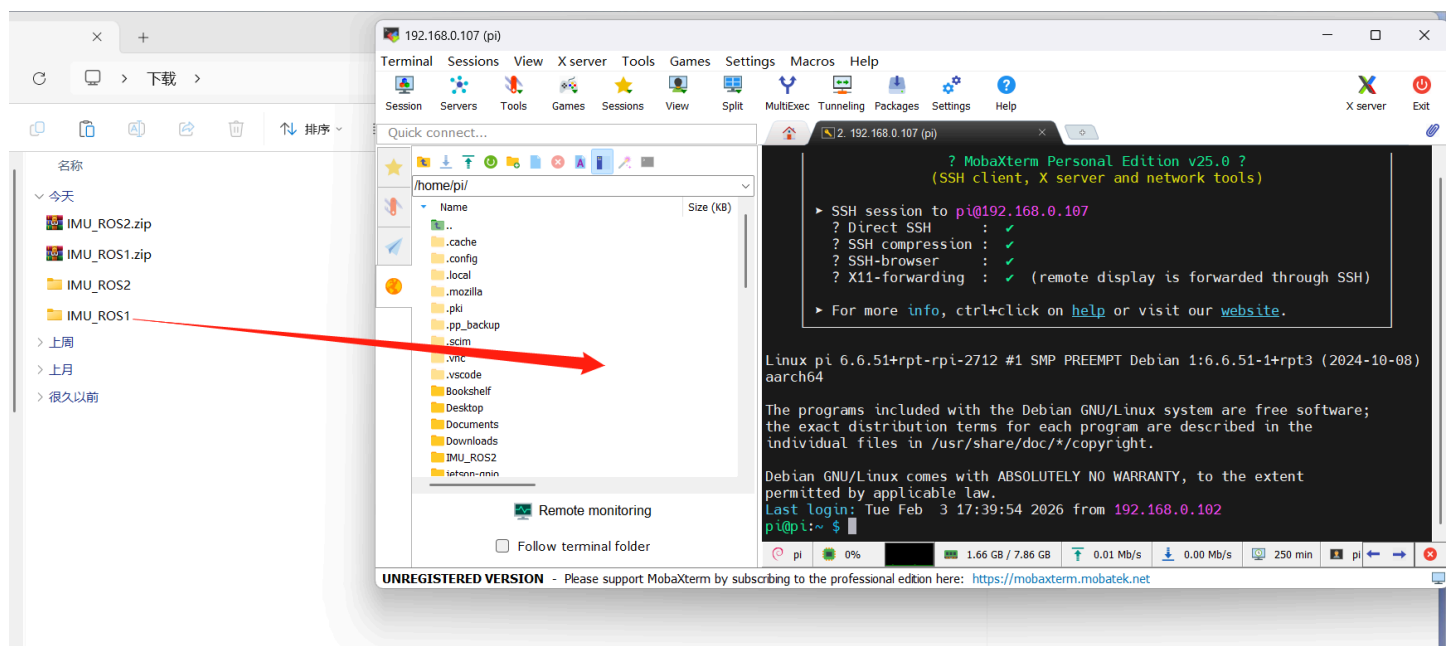
IMU_ROS2.zip

357.96KB



Friends who are not yet familiar with using MobaXterm to transfer files, please refer to the following webpage for detailed installation and operation methods of MobaXterm: [国文件远程传输](#)

Drag the decompressed files onto Raspberry Pi 5 via MobaXterm software.



4. View IMU data

Enter the ~/IMU_Library directory and run the IMU_Serial_Library.py file

代码块

```
1 cd ~/imu_ros1/src/IMU_ROS1/IMU_Library
2 # 或
3 cd ~/imu_ros2/src/IMU_ROS2/IMU_Library
4
5 # 运行 IMU 数据打印文件
6 python3 -m IMU_Library.IMU_I2C_Library
```

```
pi@pi: ~/IMU_ROS2/IMU_Library
文件(E) 编辑(E) 标签(I) 帮助(H)
pi@pi:~/IMU_ROS2/IMU_Library $ python3 -m IMU_Library.IMU_I2C_Library
<frozen runpy>:128: RuntimeWarning: 'IMU_Library.IMU_I2C_Library' found in sys.modules after import of package 'IMU_Library', but prior to execution of 'IMU_Library.IMU_I2C_Library'; this may result in unpredictable behaviour
固件版本: 1.1.2
加速度计: [-0.036133915219580676, 0.185064241462447, 0.9858699301126133]
陀螺仪: [0.0, 0.0, 0.0]
磁力计: [-18.750572222052675, 9.595019379253518, -59.98718222602008]
姿态角: [10.332492464115495, 2.0441309039517765, -126.26366929879944]
四元数: [0.44660186767578125, 0.05653122067451477, -0.07248134911060333, -0.888099193572998]
气压计: [9.31, 18.2, 100963.125, 101074.78125]

加速度计: [-0.03662221137119663, 0.18555253761406293, 0.9863582262642292]
陀螺仪: [0.0, 0.0, 0.0]
磁力计: [-12.50038148136845, 9.88799707022309, -59.54771568956572]
姿态角: [10.295379777486586, 2.0932491421508654, -126.21013427608487]
四元数: [0.4470323920249939, 0.05681021884083748, -0.07197203487157822, -0.8879090547561646]
气压计: [9.23, 18.2, 100964.03125, 101074.78125]

加速度计: [-0.036133915219580676, 0.18457594531083102, 0.9868465224158453]
陀螺仪: [0.0, 0.0, 0.0]
磁力计: [-12.744529557176428, 9.814752647480697, -59.596545304727314]
姿态角: [10.307469212317507, 2.0607922966583767, -126.18667940647505]
四元数: [0.44719696044921875, 0.056611135601997375, -0.07218325138092041, -0.8878190517425537]
气压计: [9.36, 18.21, 100962.5625, 101074.78125]
```

Attention: The above is the data reading of 10-axis IMU, 6-axis without magnetometer and barometer data, 9-axis without barometer data.

5. IMU Calibration

Enter the ~/IMU_Library directory and run the imu_calibration_tool.py file

代码块

```
1  cd ~/IMU_Library
2
3  # 运行 IMU 校准代码文件 --I2C通讯校准
4  # 执行所有校准（整体、磁力计、温度）
5  python3 -m IMU_Library.imu_calibration_tool --mode i2c --port 0
6
7  # 仅整体校准
8  python3 -m IMU_Library.imu_calibration_tool --mode i2c --port 0 --calibrate imu
9
10 # 仅磁力计校准
11 python3 -m IMU_Library.imu_calibration_tool --mode i2c --port 0 --calibrate mag
12
13 # 仅温度校准
14 python3 -m IMU_Library.imu_calibration_tool --mode i2c --port 0 --calibrate temp
```

```
pi@pi: ~/IMU_ROS2/IMU_Library
文件(F) 编辑(E) 标签(T) 帮助(H)
pi@pi:~/IMU_ROS2/IMU_Library $ python3 -m IMU_Library.imu_calibration_tool --mode i2c --port 1
=====
IMU校准工具
支持I2C和串口通信模式
=====
正在初始化I2C通信: 总线 1
Firmware version: 1.1.2

=== 开始IMU整体校准 ===
请将IMU保持静止水平放置...
IMU整体校准结果: success

=== 开始磁力计校准 ===
请缓慢旋转IMU, 确保各个方向都被测量到...
校准过程需要约10秒, 请保持旋转...
磁力计校准结果: success

=== 开始温度传感器校准 ===
请输入当前环境温度(摄氏度, 例如: 25.0): 17
正在使用温度 17.0°C 进行校准...
请保持IMU温度稳定...
温度校准结果: success

=====
校准完成!
结果: 3/3 项校准成功
✓ 所有校准项目均已成功完成!

正在关闭IMU连接...
校准工具已退出
```

6. Precautions

When using the RDK X5 main board, it is necessary to modify the serial number of the I2C bus according to the actual situation. The modification position is shown in the figure below. Usually, it is bus number 0.

```
IMU_I2C_Library.py
文件 编辑 查看
class IMU_I2C:
    """IMU传感器I2C通信类"""

    def __init__(self, port=1, debug=False):
        """
        初始化IMU I2C通信

        参数:
            port: I2C总线端口号, 默认1
            debug: 是否启用调试模式, 默认False
        """
        self._port = int(port)
        self._delay_time = 0.001 # 数据读取延迟时间
        self._debug = debug      # 调试模式标志
        self._addr = 0x23        # I2C设备地址
```

```
IMU_I2C_Library.py
文件 编辑 查看
...
if self._debug:
    print("-----I2C数据接收已停止-----")

if __name__ == '__main__':
    # 示例代码
    imu = IMU_I2C(port=1, debug=True)

    # 获取固件版本
    version = imu.get_firmware_version()
    print(f"固件版本: {version}")

    # 以下校准功能可根据需要启用
    # time.sleep(1)
    # imu.calibrate_imu()
    # imu.calibrate_magnetometer()
    # imu.calibrate_temperature(25.1)
    # imu.set_fusion_algorithm(9)
    # imu.reset_calibration_data()

    time.sleep(1)

    try:
        while True:
```